A Mathematical Framework for Transformer Circuits

AUTHORS

Nelson Elhage*†, Neel Nanda*, Catherine Olsson*, Tom Henighan†, Nicholas Joseph†, Ben Mann†, Amanda Askell, Yuntao Bai, Anna Chen, Tom Conerly, Nova DasSarma, Dawn Drain, Deep Ganguli, Zac Hatfield-Dodds, Danny Hernandez, Andy Jones, Jackson Kernion, Liane Lovitt, Kamal Ndousse, Dario Amodei, Tom Brown, Jack Clark, Jared Kaplan, Sam McCandlish, Chris Olah‡

* Core Research Contributor; † Core Infrastructure Contributor;
‡ Correspondence to colah@anthropic.com; Author contributions statement below.

AFFILIATION

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HUMANE Lab 석사과정 고경빈

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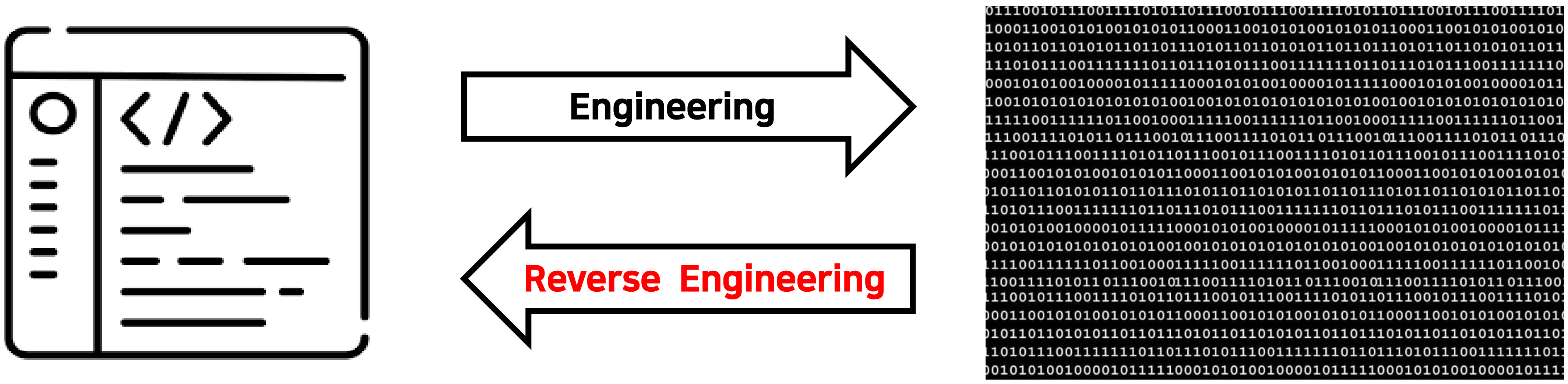
Background

- Transformer language models are an emerging technology(e.g., GPT-3, LaMDA, Codex, etc.)
- As the models scale, their open-endedness and high capacity create increasingly unexpected and harmful behaviors
- Even years after a large model is trained, previously unknown capabilities are discovered

How to address this issues?

Reverse Engineering



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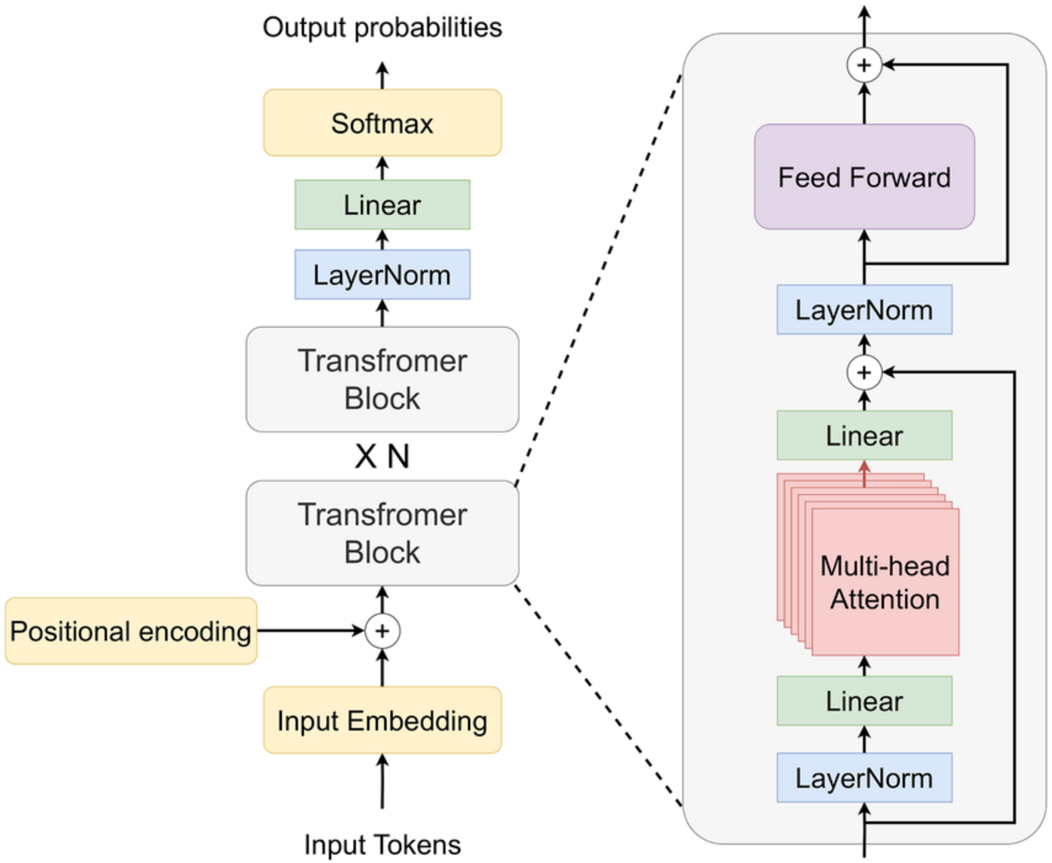
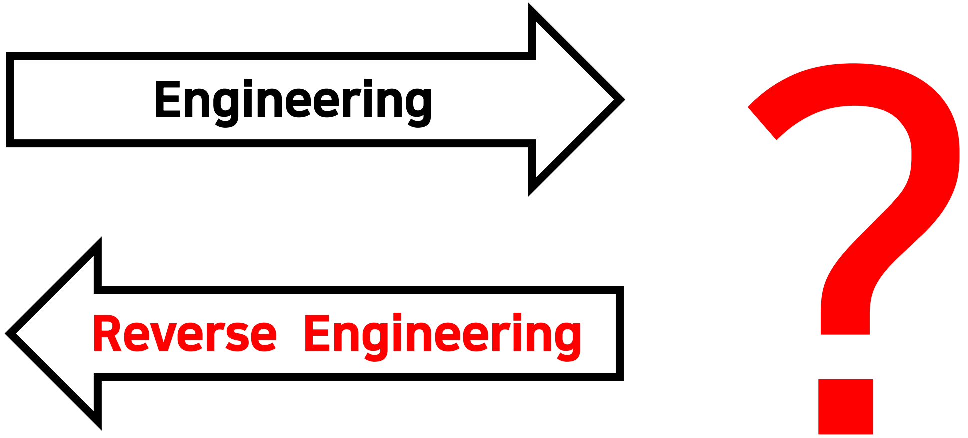
- The process of deconstructing a finished product/system to understanding it

Reverse Engineering

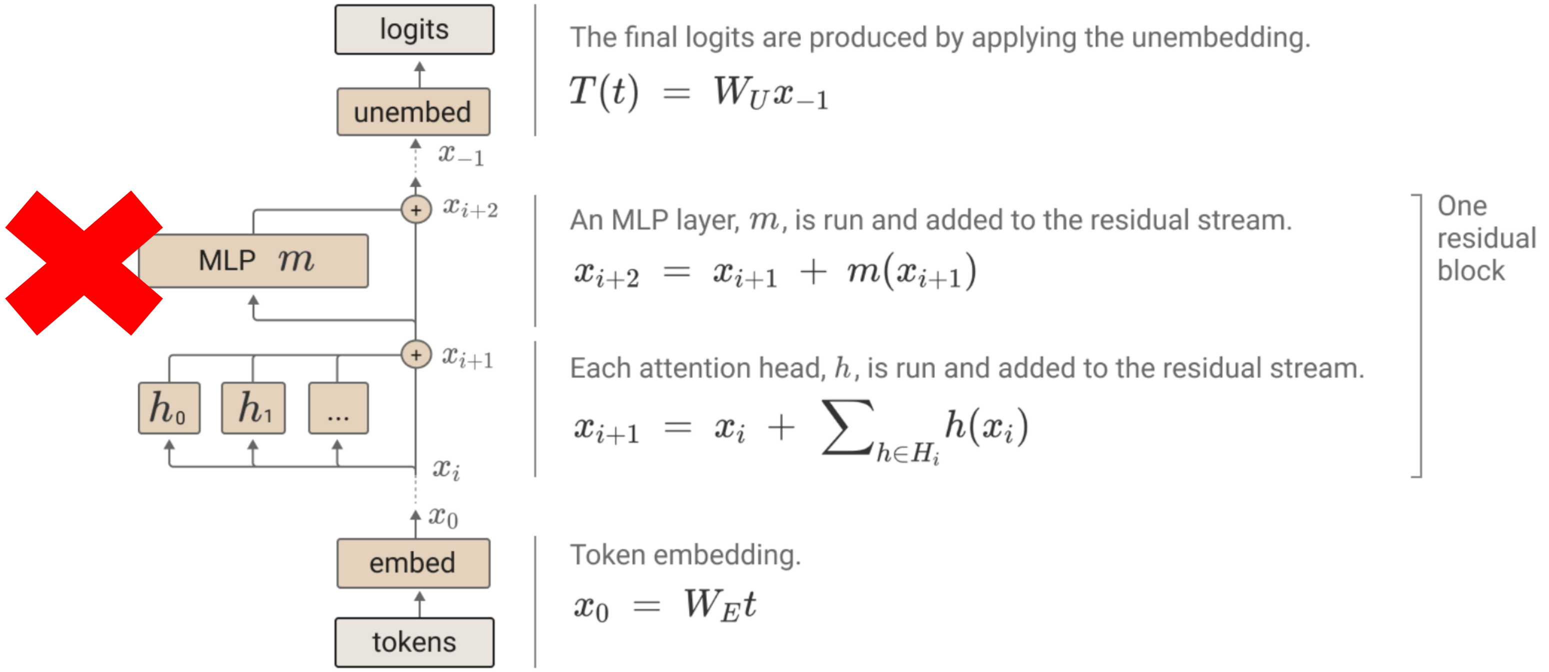


< Decoder-only Transformer >



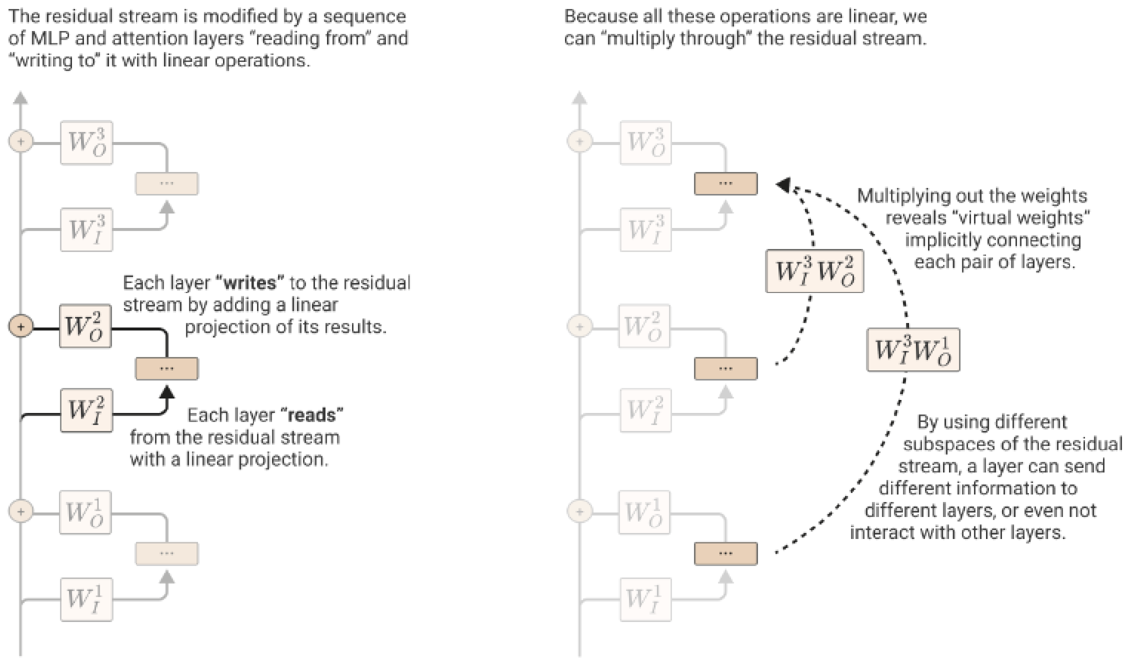
Model Simplifications

- Focus on “attention-only” transformers without MLP layers
- Do not consider biases(bias folding) and layer normalization



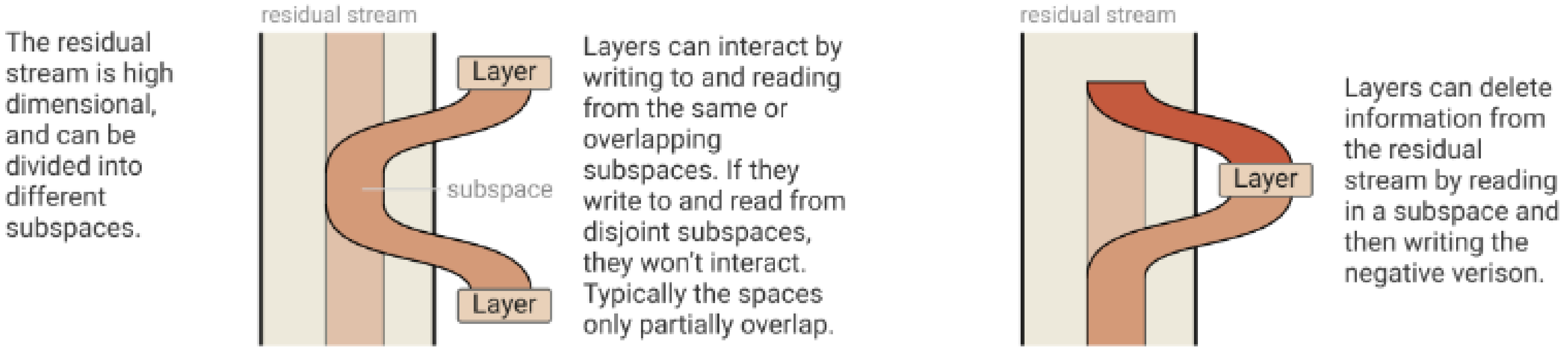
Residual Stream

- Communication Channel
  - It doesn't do any processing itself and all layers communicate through it
- Virtual Weights
  - The extent to which later layer reads in the information written by a previous layer



Residual Stream

- Subspaces for bandwidth
  - Every attention head operates on a small subspace, enabling it to write to disjoint subspaces without interaction
  - Once added, information persists in a subspace unless another layer actively deletes it



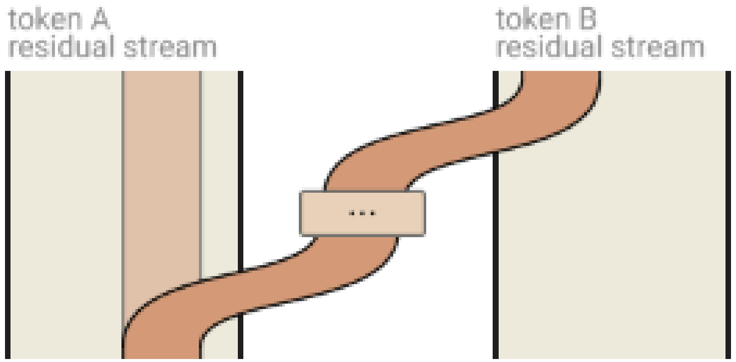
- Bottleneck Activations
  - There are generally far more computational dimensions than the residual stream has dimensions to move information
  - Focus on decomposition of the residual stream into virtual weights

Attention Heads are Independent and Additive

$$\boxed{W_O^H \begin{bmatrix} r^{h_1} \\ r^{h_2} \\ \cdots \end{bmatrix}} = \left[W_O^{h_1}, W_O^{h_2}, \ldots\right] \cdot \begin{bmatrix} r^{h_1} \\ r^{h_2} \\ \cdots \end{bmatrix} = \boxed{\sum_i W_O^{h_i} r^{h_i}}$$

- Original transformers: Stacking the result vectors(r^{h_i}) and then multiplying by an output matrix(W_O^H)
- Ours: Split output matrix(W_O^H) into equal size blocks

Attention Heads as Information Movement



Attention heads copy information from the residual stream of one token to the residual stream of another. They typically write to a different subspace than they read from.

Attention Heads as Information Movement

- Computing the output of an attention head(Original):
  1. Compute the value vector for each token from the residual stream ($v_i = W_V x_i$)
  2. Compute the result vector by linearly combining value vectors according to the attention pattern ($r_i = \sum_j A_{i,j} v_j$)
  3. Finally, compute the output vector of the head for each token ($h(x)_i = W_O r_i$)
- Using tensor products to describe mapping different operations to a matrix(Ours):

$$h(x) = \underbrace{(\mathrm{Id} \otimes W_O)}_{\substack{\text{Project result vectors out for each token} \\ (h(x)_i = W_O r_i)}} \cdot \underbrace{(A \otimes \mathrm{Id})}_{\substack{\text{Mix value vectors } \textit{across} \text{ tokens to compute result vectors} \\ (r_i = \sum_j A_{i,j} v_j)}} \cdot \underbrace{(\mathrm{Id} \otimes W_V)}_{\substack{\text{Compute value vector for each token} \\ (v_i = W_V x_i)}} \cdot x = \underbrace{(A \otimes W_O W_V)}_{\substack{A \text{ mixes across tokens while} \\ W_O W_V \text{ acts on each vector independently.}}} \cdot x$$

  - Attention Pattern(A) = softmax($\underbrace{x^T W_Q^T}_{\text{queries}} \underbrace{W_K x}_{\text{keys}}$)

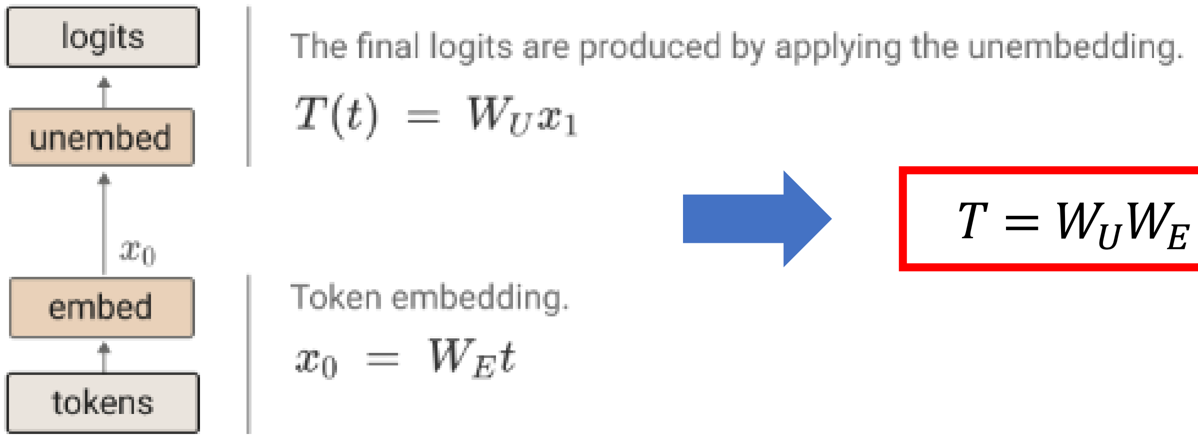
Rewriting Attention Heads reveals previously hidden structures

h(x) = $\left(A \otimes W_O W_V\right) \cdot x$

- Attention heads move information from the residual stream of one token to another
- An attention head applies two independent linear operations
    - A: Which token's information is moved from and to
    - $W_O W_V$: Which information is read from the source token and how it is written to the destination token
- Virtual Attention Heads: $\left(A^{h_2} \otimes W_{OV}^{h_2}\right) \cdot \left(A^{h_1} \otimes W_{OV}^{h_1}\right) = \left(A^{h_2} A^{h_1}\right) \otimes \left(W_{OV}^{h_2} W_{OV}^{h_1}\right)$

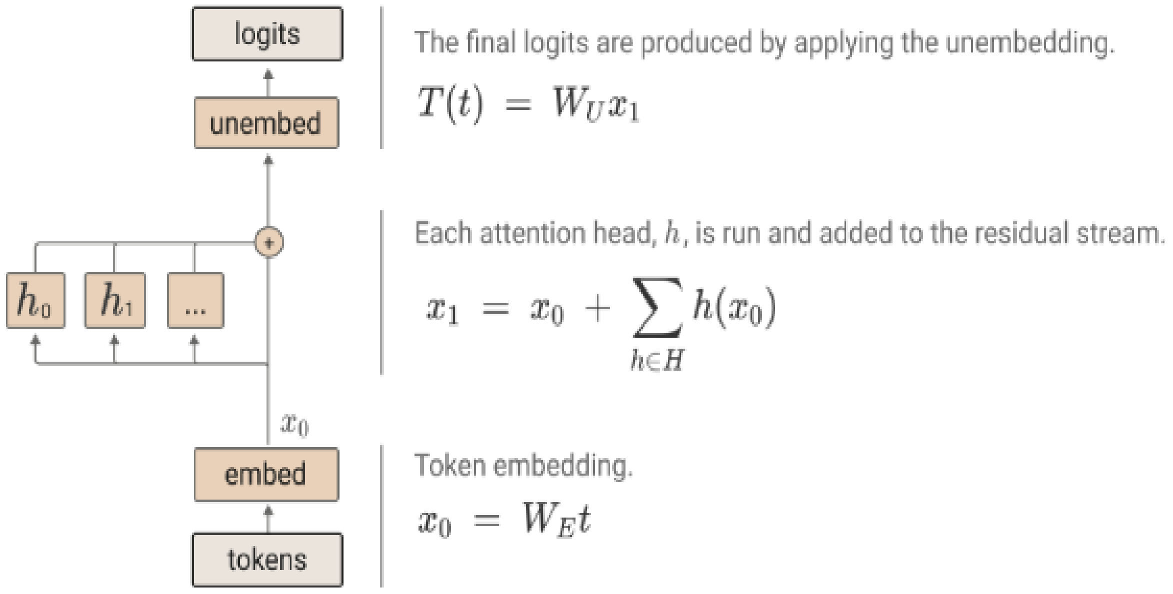
Zero-Layer Transformers

- Simply predicting the next token from the present token, since the model cannot move information
- $W_U W_E$ term seems to help represent bigram statistics which aren't described by grammatical rules
  - E.g., "Barack" is often followed by "Obama"



$$\boxed{T = W_U W_E}$$

One-Layer Attention-Only Transformers



- An ensemble of a bigram model and several skip-trigram models

One-Layer Attention-Only Transformers

$$T = \mathrm{Id} \otimes W_U \cdot \left(Id + \sum_{h \in H_1} A^h \otimes W_{OV}^h \right) \cdot \mathrm{Id} \otimes W_E = \mathrm{Id} \otimes W_U W_E + \sum_{h \in H} A^h \otimes (W_U W_{OV}^h W_E)$$

The **token unembedding** maps residual stream vectors to logits.

The **attention layer** has multiple heads. The result of each is added into the residual stream.

The **token embedding** maps tokens to residual stream vectors.

"Direct path" term contributes to bigram statistics.

The **attention head** terms describe the effects of attention heads in linking input tokens to logits. A^h describes which tokens are attended to while $W_U W_{OV}^h W_E$ describes how each token changes the logits if attended to.

$$\text{where} \quad A^h = \mathrm{softmax}^* \left(t^T \cdot W_E^T W_{QK}^h W_E \cdot t \right)$$

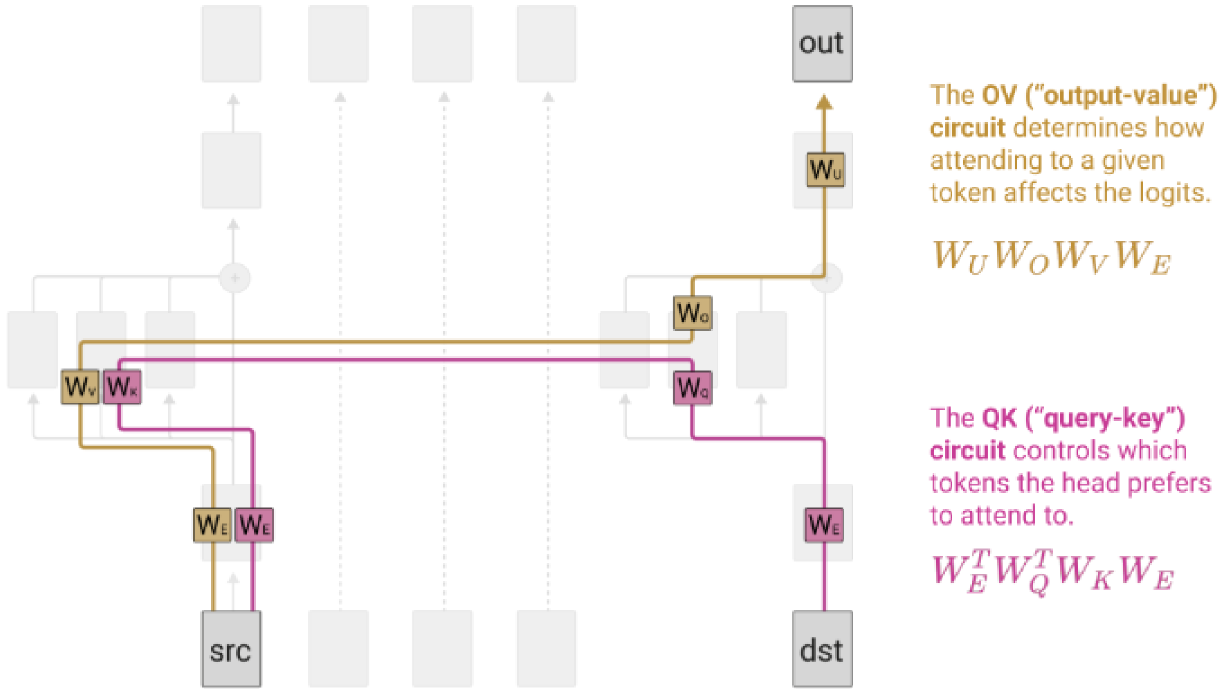
Softmax with autoregressive masking

Attention pattern logits are produced by multiplying pairs of tokens through different sides of W_{QK}^h.

- Each terms is tractable to understand, can be reasoned about independently, and combine to create model behavior

Splitting Attention Head terms into Query-Key and Output-Value Circuits

- $A^h \otimes (W_U W_{OV}^h W_E) = softmax\left(t^T \cdot \boxed{W_E^T W_{QK}^h W_E} \cdot t\right) \otimes \left(\boxed{W_U W_{OV}^h W_E}\right)$



- Once a destination token has decided how much to attend to a source token, the effect on the output is solely a function of that source token

Interpretation as Skip-Trigrams

- Skip-Trigrams: [source] ··· [destination][out]
- Most attention heads in one-layer models dedicate an enormous fraction of their capacity to copying

Some examples of large entries QK/OV circuit

Source Token	Destination Token	Out Token	Example Skip Tri-grams
" perfect"	" are", " looks", " is", " provides"	" perfect", " super", " absolute", " pure"	" perfect... are perfect", " perfect... looks super"
"lambda"	" $\\", "}{\\", "+\\", "(\\", " ${\\"	"lambda", "sorted", " lambda", "operator"	"lambda... $\\lambda", "lambda... +\\lambda"
"nbsp"	"&", " \"&", "}&", ">&", "=&"	"nbsp", "01", "gt", "00012", "nbs", "quot"	"nbsp... ", "nbsp... > "
"Great"	"The", " The", " the", " contains", " /"	" Great", " great", " poor", " Every"	"Great... The Great", "Great... the great"

Source Token	Destination Token	Out Token	Example Skip Tri-grams
"indy"	" C", "C", " V", "V", " R", " c"	"indy", "obby", "INDY", "loyd"	"indy... Cindy", "indy... CINDY"
" Pike"	" P", "P", "V", "Sp", " V", "R"	"ike", "ikes", "ishing", "owler"	" Pike... Pike", " Pike... Spikes"
" Ralph"	"R", " R", " P", "P", "V", " r"	"alph", "ALPH", "obby", "erald"	" Ralph... Ralph", " Ralph... RALPH"
" Lloyd"	"L", " L", " P", "P", "R", " C"	"loyd", "alph", "\n ", "acman", ... "atherine"	" Lloyd... Lloyd", " Lloyd... Catherine"

Primitive In-Context Learning Patterns

[b]...[a]→[b]	[b]...[a]→[b']	[ab]...[a]→[b]	[ab]...[a]→[b']
[two]...[One]→[two]	[two]...[has]→[three]	[Ralph]...[R]→[alph]	[Ralph]...[R]→[ALPH]
[perfect]...[are]→[perfect]	[perfect]...[looks]→[super]	[Pike]...[P]→[ike]	[Pike]...[P]→[ikes]
[nbsp]...[&]→[nbsp]	[nbsp]...[&]→[gt]	[Pixmap]...[P]→[ixmap]	
[lambda]...[$\\]→[lambda]	[lambda]...[$\\]→[operator]	[Lloyd]...[L]→[loyd]	

Interpretation as Skip-Trigrams

- Skip-Trigram Bugs: $f(a, b, c) = f_1(a, b) f_2(a, c)$ can't capture the three way interactions flexibly
  - If a single head increases the probability of both "keep ⋯ in mind" and "keep ⋯ at bay", it must also increase the probability of "keep ⋯ in bay" and "keep ⋯ at mind"

Limited Expressivity Can Create Bugs which Seem Strange from the Outside

Source Token	Destination Token	Out Token	"Correct" Skip Tri-grams	"Bug" Skip Tri-grams
" Pixmap"	" P", " Q", "P", " p", " U"	"ixmap", "Canvas", "Embed", "grade"	" Pixmap... Pixmap", " Pixmap... QCanvas"	" Pixmap... PCanvas"

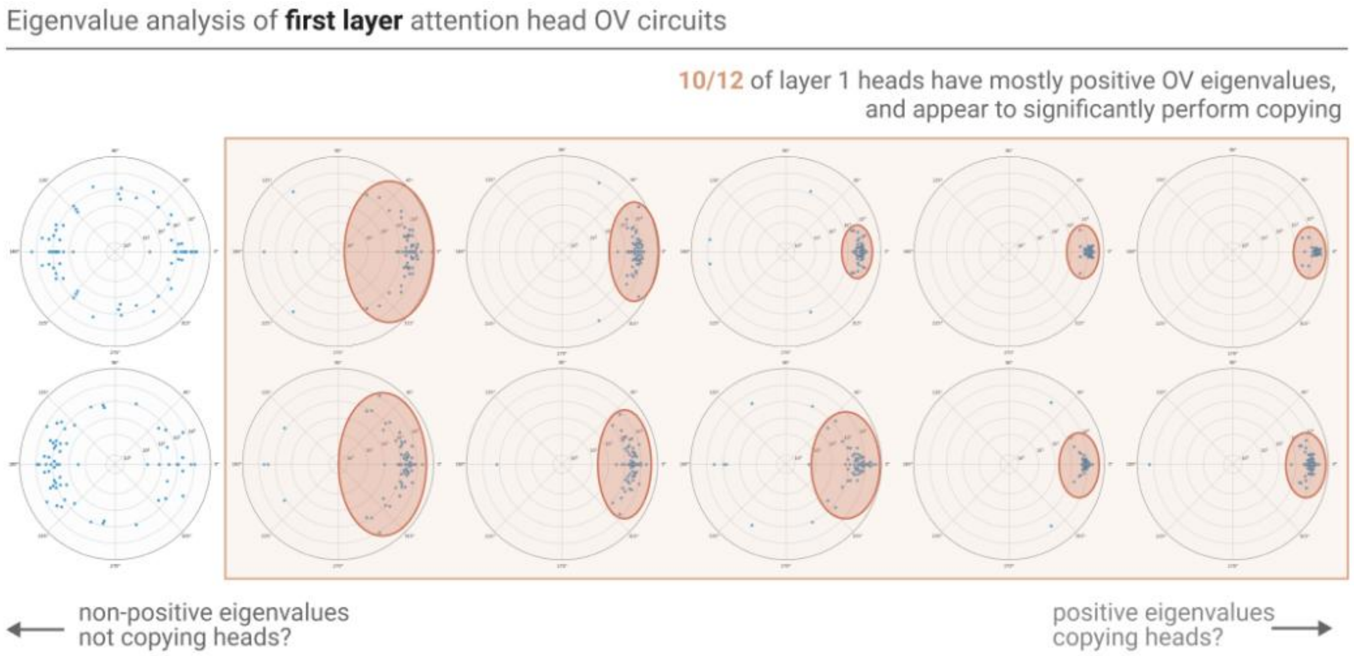
Source Token	Destination Token	Out Token	"Correct" Skip Tri-grams	"Bug" Skip Tri-grams
" Lloyd"	"L", " L", " P", "P", "R", " C"	"loyd", "alph", "\n ", "acman", ... "atherine"	" Lloyd... Lloyd", " Lloyd... Catherine"	" Lloyd... Cloyd", " Lloyd... Latherine"

Source Token	Destination Token	Out Token	"Correct" Skip Tri-grams	"Bug" Skip Tri-grams
" keep"	" in", " at", " out", " under", " off"	" bay", ... " mind", ... " wraps"	" keep... in mind", " keep... at bay", " keep... under wraps"	" keep... in bay", " keep... at wraps", " keep... under mind"

▶ Shed light on transformer behavior that seems incomprehensible from the outside

Summarizing OV/QK Matrices

- The expanded OV and QK matrices are so enormous that it is impossible to inspect every value
- Using eigenvectors and eigenvalues($Mv_i = \lambda_i v_i$)
  - Copy: Mapping the same vector to itself, increasing its possibility
  - If λ_i is a positive real number, OV circuit($M = W_U W_{OV}^h W_E$) means that the information is preserved and its value is amplified



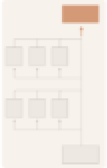
Two-Layer Attention-Only Transformers

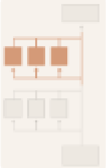
- Attention heads are composed through Q-, K-, and V-Composition
- Q- and K-Composition are quite different from V-Composition
  - Q- and K-Composition: Affect the attention pattern, allowing attention heads to express much more complex patterns
  - V-Composition: Affects what information an attention head moves when it attends to a given position

- To understand these three kinds of composition, we'll need to study the OV and QK circuits again ...

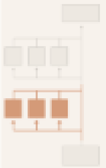
Path Expansion of Logits

$$T = \mathrm{Id} \otimes W_U \cdot \left(Id + \sum_{h \in H_2} A^h \otimes W_{OV}^h\right) \cdot \left(Id + \sum_{h \in H_1} A^h \otimes W_{OV}^h\right) \cdot \mathrm{Id} \otimes W_E$$

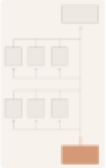




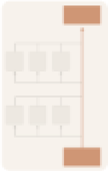
The second **attention layer** has multiple attention heads which add into the residual stream



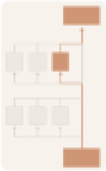
The first **attention layer** has multiple attention heads which add into the residual stream



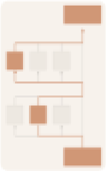
$$= \mathrm{Id} \otimes W_U W_E + \sum_{h \in H_1 \cup H_2} A^h \otimes (W_U W_{OV}^h W_E) + \sum_{h_2 \in H_2} \sum_{h_1 \in H_1} (A^{h_2} A^{h_1}) \otimes (W_U W_{OV}^{h_2} W_{OV}^{h_1} W_E)$$



"Direct path" term contributes to bigram statistics.



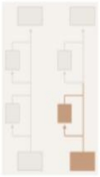
The **individual attention head** terms describe the effects of individual attention heads in linking input tokens to logits, similar to those we saw in the one layer model.



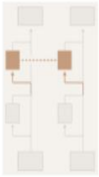
The **virtual attention head** terms correspond to V-composition of attention heads. They function a lot like individual attention heads, with their own attention patterns (the compositon of the heads patterns) and own OV matrix.

Path Expansion of Attention Scores QK Circuit

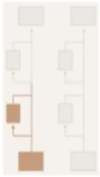
$$C_{QK}^{h\in H_2} = x_1^T W_{QK}^h x_1 = \left(\mathrm{Id}\otimes\mathrm{Id}\otimes W_E^T + \sum_{h_q\in H_1} A^{h_q}\otimes\mathrm{Id}\otimes(W_{OV}^{h_q}W_E)^T\right) \cdot \mathrm{Id}\otimes\mathrm{Id}\otimes W_{QK}^h \cdot \left(\mathrm{Id}\otimes\mathrm{Id}\otimes W_E + \sum_{h_k\in H_1} \mathrm{Id}\otimes A^{h_k}\otimes W_{OV}^{h_k}W_E\right)$$



The **"query side" residual stream** at the start of the second layer contains both the layer 1 direct path and layer 1 attention heads. All terms are of the form $\ldots\otimes\mathrm{Id}\otimes\ldots$ because they don't move key information.



W_{QK} of the second layer head combines both sides into attention scores.



The **"key side" residual stream** at the start of the second layer contains both the layer 1 direct path and attention heads. All terms are of the form $\mathrm{Id}\otimes\ldots$ because they don't move query information.

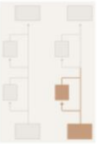
- Problem: We have to write it as a 6-dimensional tensor, using two tensor products on matrices

→ Each term will be the of the form $A_q \otimes A_k \otimes W$ where $x(A_q \otimes A_k \otimes W)y = A_q^T x W y A_k$

- A_q: The movement of query-side information between tokens
- A_k: The movement of key-side information between tokens
- W: How they product together to form an attention score

Path Expansion of Attention Scores QK Circuit

$$C_{QK}^{h\in H_2} = \left(\mathrm{Id}\otimes\mathrm{Id}\otimes W_E^T + \sum_{h_q\in H_1} A^{h_q}\otimes \mathrm{Id}\otimes (W_{OV}^{h_q}W_E)^T\right) \cdot \mathrm{Id}\otimes\mathrm{Id}\otimes W_{QK}^{h} \cdot \left(\mathrm{Id}\otimes\mathrm{Id}\otimes W_E + \sum_{h_k\in H_1}\mathrm{Id}\otimes A^{h_k}\otimes W_{OV}^{h_k}W_E\right)$$



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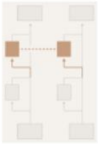
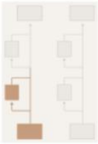
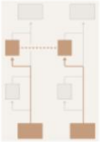


W_{QK} of the second layer head combines both sides into attention scores.

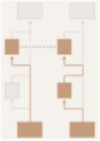


The **"key side" residual stream** at the start of the second layer contains both the layer 1 direct path and attention heads. All terms are of the form $\mathrm{Id}\otimes\ldots$ because they don't move query information.

$$= \mathrm{Id}\otimes\mathrm{Id}\otimes\left(W_E^T W_{QK}^h W_E\right) + \sum_{h_q\in H_1} A^{h_q}\otimes\mathrm{Id}\otimes\left(W_E^T W_{OV}^{h_q\,T} W_{QK}^h W_E\right)$$

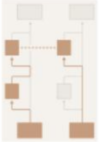


The no composition term. Both first layer follows the direct path on both the key and query side.

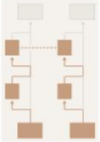


These terms correspond to pure **Q-composition**. A previous attention head is used to generate the query side, but the key side is the first layer direct path.

$$+ \sum_{h_k\in H_1} \mathrm{Id}\otimes A^{h_k}\otimes\left(W_E^T W_{QK}^h W_{OV}^{h_k} W_E\right) + \sum_{h_q\in H_1}\sum_{h_k\in H_1} A^{h_q}\otimes A^{h_k}\otimes\left(W_E^T W_{OV}^{h_q\,T} W_{QK}^h W_{OV}^{h_k} W_E\right)$$



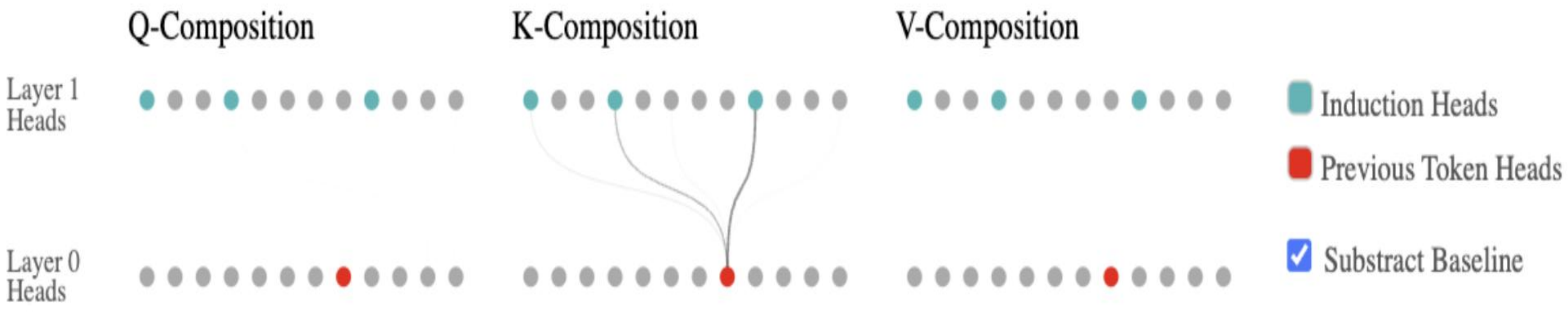
These terms correspond to pure **K-composition**. A previous attention head is used to generate part of the key, but the query side is the first layer direct path.



These terms are interactions between both **Q-composition** and **K-composition**. A previous attention head is used to generate the query and key sides.

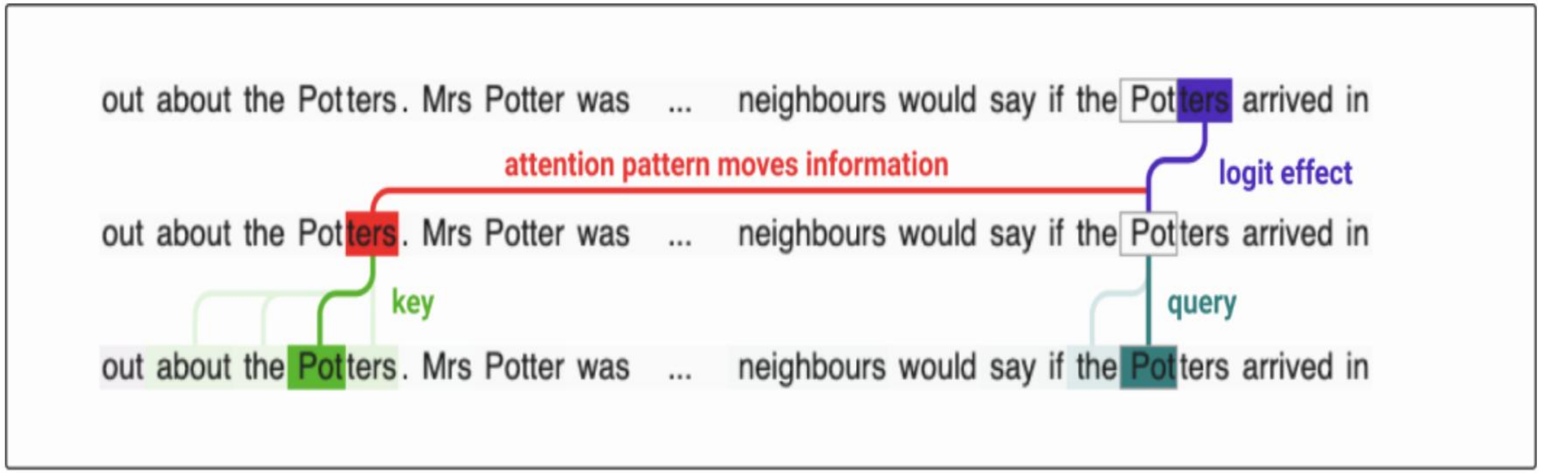
Analyzing a Two-Layer Model

- Without Q-, K-, and V-composition, two-layer model is just a one-layer model with extra heads
- Using the Frobenius norm, we can measure which heads are interacting with each other
  - Q-Composition: $\left\|W_{QK}^{h_2\,T} W_{OV}^{h_1}\right\|_F / \left(\left\|W_{QK}^{h_2\,T}\right\|_F \left\|W_{OV}^{h_1}\right\|_F\right)$
  - K-Composition: $\left\|W_{QK}^{h_2} W_{OV}^{h_1}\right\|_F / \left(\left\|W_{QK}^{h_2}\right\|_F \left\|W_{OV}^{h_1}\right\|_F\right)$
  - V-Composition: $\left\|W_{OV}^{h_2} W_{OV}^{h_1}\right\|_F / \left(\left\|W_{OV}^{h_2}\right\|_F \left\|W_{OV}^{h_1}\right\|_F\right)$



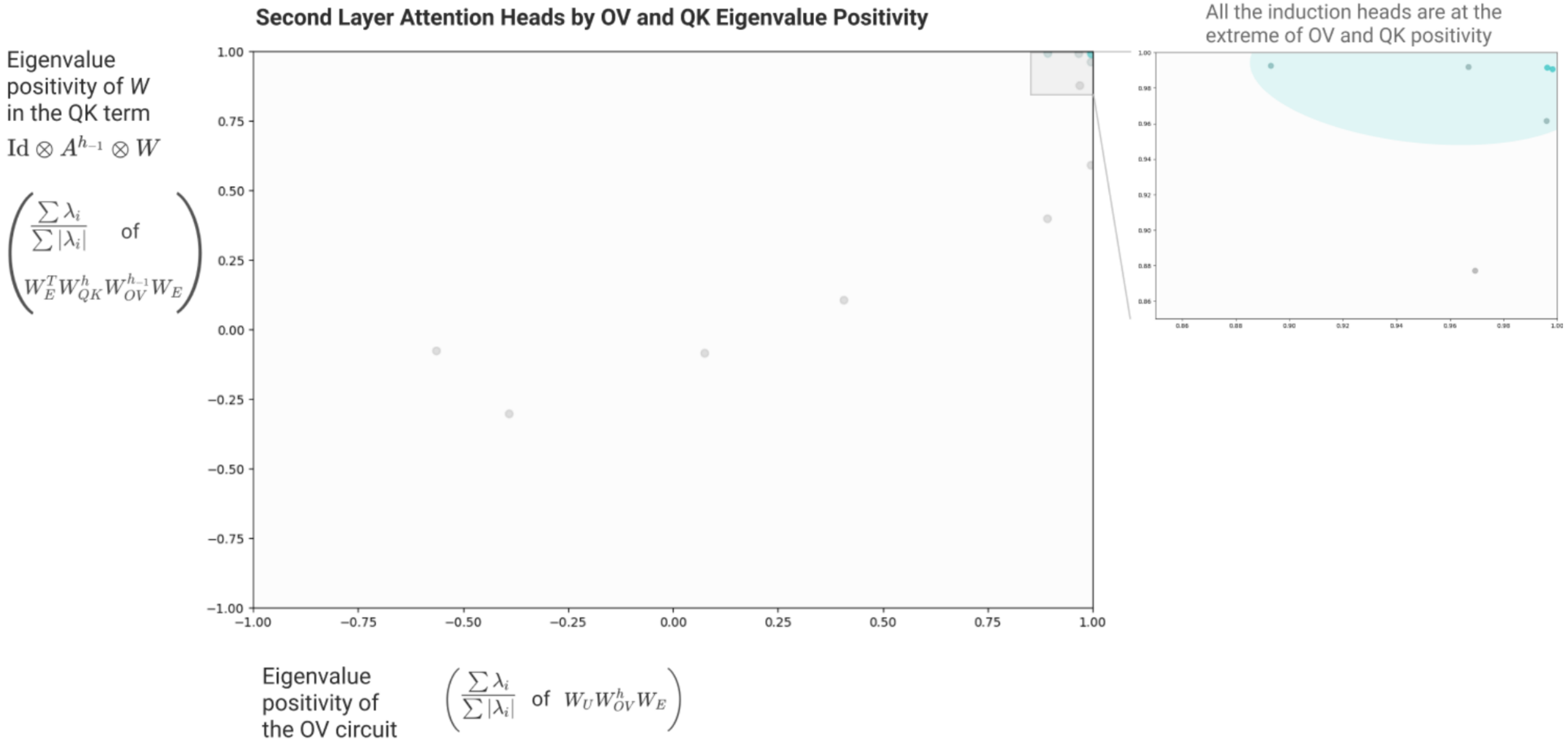
Induction Heads

- Function of Induction Heads
  - One layer model copying head: [b] ··· [a] → [b]
  - Two layer model induction head: [a][b] ··· [a] → [b]
- How Induction Heads Work
  - The central trick to induction heads is that the key is computed from tokens shifted one token back
  - QK circuit = $Id \otimes A^{h-1} \otimes W$



Checking the Mechanistic Theory of Induction Head

- Our mechanistic theory suggestions that induction heads must do two things:
  - Have a "copying" OV circuit matrix
  - Have a "same matching" QK circuit matrix associated with the $Id \otimes A^{h_{-1}} \otimes W$ term



Conclusion

- Transformers can be mathematically reverse-engineered
- Deeper models develop new reasoning skills beyond simple memorization
- Induction heads are the core engine driving LLMs' context learning

My Opinion

- Transformer 수식을 조금씩 변형하여 Reverse Engineering 한 것이 가장 큰 기여점인 것 같다(분석 결과는 다른 연구에서 이미 나온 것)
- 모델을 너무 단순화한 것이 단점인 것 같다. 실제 Transformer 구조와 차이가 크다 보니 이게 진짜 Transformer의 Reverse Engineering이라고 할 수 있을까?